

**B.E. Electrical Engineering Second Year Second
Semester Examination 2025
ELECTRICAL INSTRUMENTATION PART – II**

Examination Solutions

Q1. Explain the functions of “Trigger Input” in a Cathode Ray Oscilloscope (CRO). [5]

Answer

In a Cathode Ray Oscilloscope (CRO), the **trigger input** is used to synchronise the **horizontal time-base sweep** with the **vertical input signal**. Its main function is to start each sweep of the electron beam from the same point of the input waveform. Hence, a periodic waveform appears steady and stationary on the screen instead of moving or rolling.

In a CRO, the input signal is applied to the vertical amplifier. A small portion of this signal is also taken to the trigger circuit. The trigger circuit compares the signal voltage with a selected trigger level. When the signal crosses this level with the selected slope, either positive-going or negative-going, the trigger circuit produces a trigger pulse. This pulse starts the time-base generator, which produces the saw-tooth voltage for horizontal deflection.

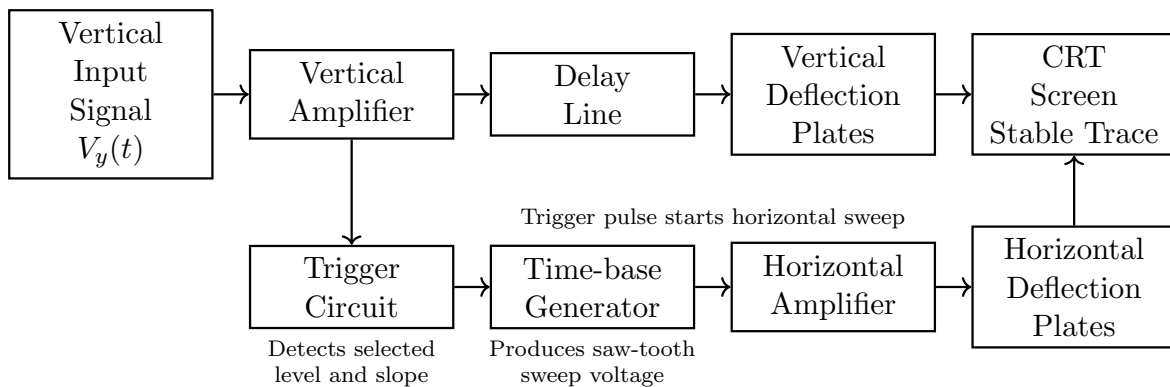


Figure 1: Labeled block diagram showing the function of trigger input in CRO.

Functions of Trigger Input

- 1. Synchronisation:** The trigger input synchronises the horizontal sweep voltage with the vertical input signal.

2. **Stable display:** It makes the waveform stationary on the CRO screen by starting every sweep at the same point of the waveform.
3. **Trigger level selection:** It allows the sweep to begin only when the input signal reaches a selected voltage level.
4. **Slope selection:** It allows triggering either on the rising edge or on the falling edge of the signal.
5. **Prevention of waveform rolling:** Without proper triggering, the waveform may drift or roll continuously on the screen. Triggering prevents this and gives a clear display.
6. **External synchronisation:** In external trigger mode, a separate external signal may be applied to the trigger input to synchronise the CRO with another circuit or reference signal.
7. **Accurate measurement:** Since the waveform starts from a fixed point, accurate measurement of time period, frequency and phase becomes possible.

Conclusion

Thus, the trigger input is an essential part of a CRO because it controls the starting instant of the time-base sweep. It ensures that each sweep begins at the same voltage level and same slope of the input signal, producing a stable and clear waveform on the screen.

Q1. (OR) The input resistance and capacitance of a Cathode Ray Oscilloscope (CRO) are $1.0\text{ M}\Omega$ and $2.0\text{ }\mu\text{F}$. What is the matched capacitance of the probe for proper compensation assuming $100\text{ k}\Omega$ probe resistance? Explain necessary formula. [5]

Answer

A CRO probe and the CRO input form a frequency dependent voltage divider. If the probe is not properly compensated, different frequency components of the input signal are attenuated by different amounts. As a result, the displayed waveform becomes distorted.

For proper compensation, the time constant of the probe section must be equal to the time constant of the CRO input section.

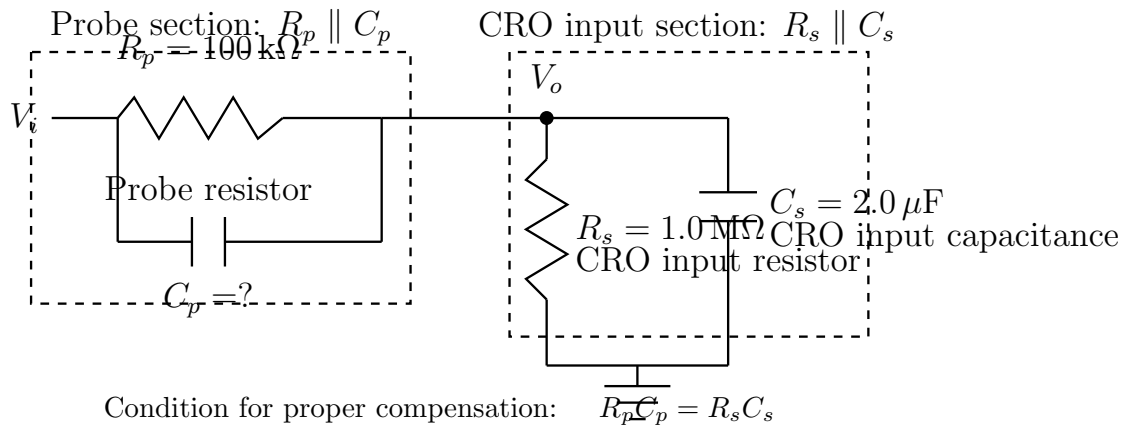


Figure 2: Equivalent circuit of compensated CRO probe showing probe resistor clearly.

Necessary Formula

Let,

$$R_p = \text{probe resistance,} \quad C_p = \text{probe capacitance}$$

$$R_s = \text{CRO input resistance,} \quad C_s = \text{CRO input capacitance}$$

For proper compensation,

$$R_s X_{C_p} = R_p X_{C_s}$$

Now,

$$X_C = \frac{1}{\omega C}$$

Therefore,

$$R_s \left(\frac{1}{\omega C_p} \right) = R_p \left(\frac{1}{\omega C_s} \right)$$

Cancelling ω ,

$$\frac{R_s}{C_p} = \frac{R_p}{C_s}$$

Hence,

$$R_s C_s = R_p C_p$$

So,

$$C_p = \frac{R_s C_s}{R_p}$$

Given Data

$$R_s = 1.0 \text{ M}\Omega = 1.0 \times 10^6 \Omega$$

$$C_s = 2.0 \mu\text{F} = 2.0 \times 10^{-6} \text{ F}$$

$$R_p = 100 \text{ k}\Omega = 100 \times 10^3 \Omega = 1.0 \times 10^5 \Omega$$

Calculation

$$C_p = \frac{R_s C_s}{R_p}$$

$$C_p = \frac{(1.0 \times 10^6)(2.0 \times 10^{-6})}{1.0 \times 10^5}$$

$$C_p = \frac{2.0}{1.0 \times 10^5}$$

$$C_p = 2.0 \times 10^{-5} \text{ F}$$

$$C_p = 20 \times 10^{-6} \text{ F}$$

$$\boxed{C_p = 20 \mu\text{F}}$$

Final Answer

Therefore, the matched capacitance of the probe for proper compensation is

$$\boxed{20 \mu\text{F}}$$

Thus, when the probe capacitance is adjusted to $20 \mu\text{F}$, the probe-CRO combination becomes properly compensated and the waveform is displayed without frequency distortion.

Solution to Question 2.(a)

Problem Statement: Draw the Switched capacitor implementation of the given active filter circuit. Assume the switching frequency is 1 kHz.

1. Mathematical Derivation of Equivalent Capacitances

In a switched-capacitor (SC) circuit, continuous-time resistors are replaced by switched capacitors toggling at a high frequency (f_s). The equivalent resistance R_{eq} of a capacitor C switched between two nodes at frequency f_s is given by the foundational formula:

$$R_{eq} = \frac{1}{f_s \cdot C} \implies C = \frac{1}{R_{eq} \cdot f_s}$$

Given the switching frequency $f_s = 1 \text{ kHz} = 1000 \text{ Hz}$, we calculate the required equivalent switched capacitors (C_{R1} and C_{R2}) to replace the standard resistors R_1 and R_2 .

Calculating C_{R1} (Replacing R_1):

$$\begin{aligned} R_1 &= 8 \text{ k}\Omega = 8 \times 10^3 \Omega \\ C_{R1} &= \frac{1}{(8 \times 10^3 \Omega) \cdot (10^3 \text{ Hz})} \\ C_{R1} &= \frac{1}{8 \times 10^6} = 0.125 \times 10^{-6} \text{ F} = \mathbf{0.125 \mu\text{F}} \end{aligned}$$

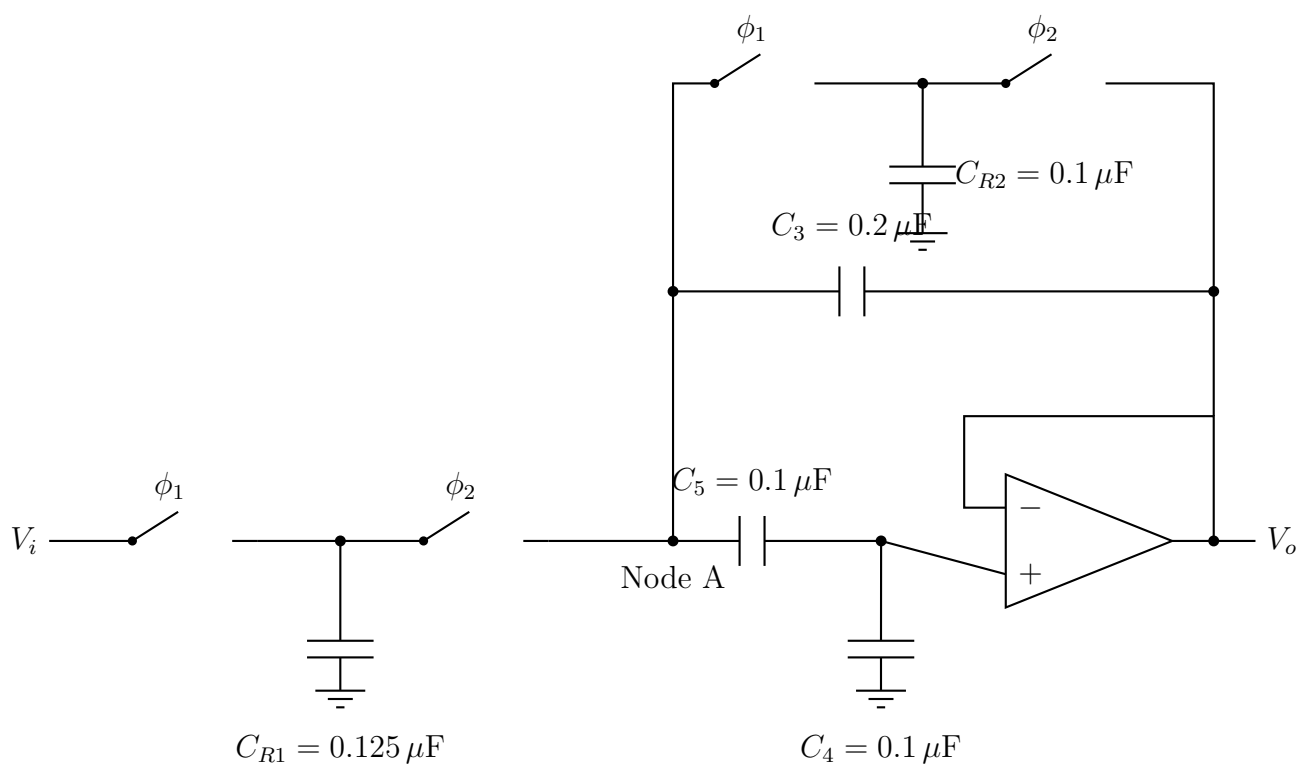
Calculating C_{R2} (Replacing R_2):

$$\begin{aligned} R_2 &= 10 \text{ k}\Omega = 10 \times 10^3 \Omega \\ C_{R2} &= \frac{1}{(10 \times 10^3 \Omega) \cdot (10^3 \text{ Hz})} \\ C_{R2} &= \frac{1}{10 \times 10^6} = 0.1 \times 10^{-6} \text{ F} = \mathbf{0.1 \mu\text{F}} \end{aligned}$$

The original continuous-time capacitors C_3 , C_4 , and C_5 remain completely unchanged in the switched-capacitor implementation:

- $C_3 = 0.2 \mu\text{F}$
- $C_4 = 0.1 \mu\text{F}$
- $C_5 = 0.1 \mu\text{F}$

Switched Capacitor Circuit Implementation



Solution to Question 2. (b)

Problem Statement: Why State Variable Filters are called Universal Filter? How can you realize a 2nd order bandpass filter using state variable representation? Show necessary derivations.

1. Why State Variable Filters are called Universal Filters

A **state variable filter** is an active filter which is realised using integrators and a summing amplifier. It is called a **universal filter** because from the same circuit, different outputs can be obtained by tapping the signal at different points.

From a single state variable filter circuit, we can obtain:

1. Low-pass filter output,
2. High-pass filter output,
3. Band-pass filter output,
4. Band-stop or notch filter output.

Thus, all major second order filter responses can be obtained from one basic circuit. Therefore, a state variable filter is also known as a **universal active filter**.

2. General Second Order Transfer Function

The general second order transfer function may be written as

$$H(s) = \frac{a_2 s^2 + a_1 s + a_0}{s^2 + b_1 s + b_0}$$

Here,

$$a_2 s^2 \rightarrow \text{high-pass term}$$

$$a_1 s \rightarrow \text{band-pass term}$$

$$a_0 \rightarrow \text{low-pass term}$$

So, the same denominator can produce high-pass, band-pass and low-pass responses depending on the numerator.

3. State Variable Representation

For a second order state variable filter, the transfer functions are written as

$$H_{HP}(s) = \frac{K_1 s^2}{s^2 + K_2 K_4 s + K_2 K_3}$$

$$H_{BP}(s) = \frac{-K_1 K_2 s}{s^2 + K_2 K_4 s + K_2 K_3}$$

$$H_{LP}(s) = \frac{K_1 K_2 K_3}{s^2 + K_2 K_4 s + K_2 K_3}$$

The negative sign in the band-pass output appears because an inverting integrator is used. For magnitude response, the sign does not affect the band-pass nature.

4. Labeled Block Diagram of State Variable Filter

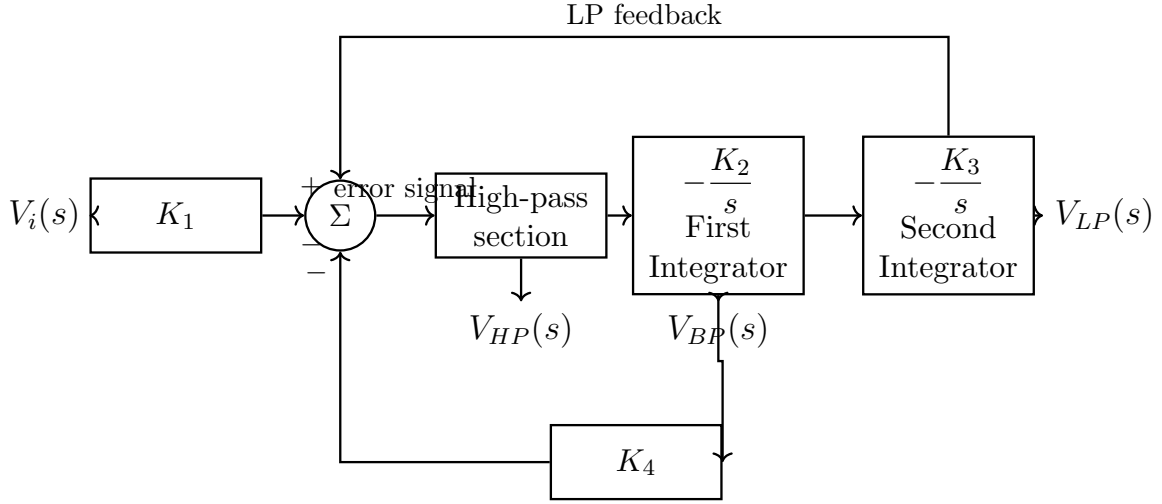


Figure 3: State variable representation showing HP, BP and LP outputs from one circuit.

5. Derivation of Second Order Band-pass Filter

Let the low-pass output be

$$V_{LP}(s) = X_1(s)$$

From the state variable structure,

$$\frac{X_1(s)}{V_i(s)} = \frac{K_1 K_2 K_3}{s^2 + K_2 K_4 s + K_2 K_3}$$

Therefore,

$$V_{LP}(s) = \frac{K_1 K_2 K_3}{s^2 + K_2 K_4 s + K_2 K_3} V_i(s)$$

The band-pass output is taken from the output of the first integrator.

Since the low-pass output is obtained after the second integrator, the band-pass output is related to the low-pass output by

$$V_{BP}(s) = -\frac{s}{K_3} V_{LP}(s)$$

Substituting the value of $V_{LP}(s)$,

$$V_{BP}(s) = -\frac{s}{K_3} \left[\frac{K_1 K_2 K_3}{s^2 + K_2 K_4 s + K_2 K_3} \right] V_i(s)$$

Cancelling K_3 ,

$$V_{BP}(s) = \frac{-K_1 K_2 s}{s^2 + K_2 K_4 s + K_2 K_3} V_i(s)$$

Hence, the transfer function of the second order band-pass filter is

$$H_{BP}(s) = \frac{V_{BP}(s)}{V_i(s)} = \frac{-K_1 K_2 s}{s^2 + K_2 K_4 s + K_2 K_3}$$

Ignoring the negative sign for magnitude response,

$$|H_{BP}(s)| = \frac{K_1 K_2 s}{s^2 + K_2 K_4 s + K_2 K_3}$$

6. Comparison with Standard Band-pass Transfer Function

The standard second order band-pass transfer function is

$$H_{BP}(s) = \frac{K_B s}{s^2 + \left(\frac{\omega_0}{Q}\right) s + \omega_0^2}$$

Comparing with

$$H_{BP}(s) = \frac{-K_1 K_2 s}{s^2 + K_2 K_4 s + K_2 K_3}$$

we get,

$$\omega_0^2 = K_2 K_3$$

$$\omega_0 = \sqrt{K_2 K_3}$$

Also,

$$\frac{\omega_0}{Q} = K_2 K_4$$

Therefore,

$$B = \frac{\omega_0}{Q} = K_2 K_4$$

and

$$Q = \frac{\omega_0}{B} = \frac{\sqrt{K_2 K_3}}{K_2 K_4}$$

7. Realisation of Band-pass Filter

To realise a second order band-pass filter using state variable representation:

1. Use one summing amplifier.
2. Use two cascaded integrators.
3. Take the output after the first integrator as the band-pass output.
4. Feed back the required signals to the summer to form the second order denominator.

The realised band-pass output is

$$V_{BP}(s) = \frac{-K_1 K_2 s}{s^2 + K_2 K_4 s + K_2 K_3} V_i(s)$$

8. Final Answer

State variable filters are called universal filters because the same circuit can produce low-pass, high-pass, band-pass and band-stop responses by taking output from different points of the circuit.

For the second order band-pass filter,

$$H_{BP}(s) = \frac{-K_1 K_2 s}{s^2 + K_2 K_4 s + K_2 K_3}$$

where,

$$\omega_0 = \sqrt{K_2 K_3}$$

$$B = K_2 K_4$$

$$Q = \frac{\sqrt{K_2 K_3}}{K_2 K_4}$$

Solution to Question 2. (OR) (a)

Problem Statement: Develop a linear model of Phase Locked Loop (PLL).

Answer

A **Phase Locked Loop (PLL)** is a feedback system in which the phase and frequency of the output signal of a voltage controlled oscillator (VCO) are locked with the phase and frequency of the input signal. A PLL mainly consists of:

1. Phase detector or phase comparator,
2. Low-pass loop filter,

3. Voltage controlled oscillator (VCO),
4. Feedback path.

1. Basic Block Diagram of PLL

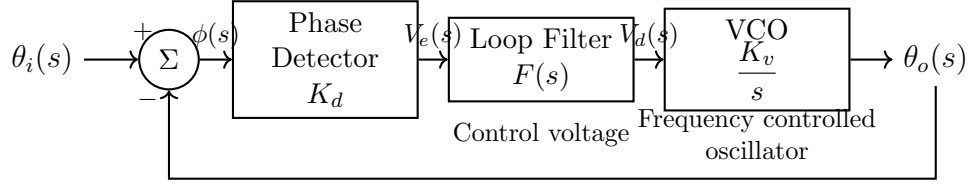


Figure 4: Linear block model of a phase locked loop.

2. Phase Error

Let,

$$\theta_i(s) = \text{input phase}$$

$$\theta_o(s) = \text{output phase of VCO}$$

The phase error is

$$\phi(s) = \theta_i(s) - \theta_o(s)$$

3. Phase Detector Output

For small phase error, the phase detector characteristic is assumed to be linear.

$$V_e(s) = K_d \phi(s)$$

where,

$$K_d = \text{phase detector gain in volt/radian}$$

Therefore,

$$V_e(s) = K_d[\theta_i(s) - \theta_o(s)]$$

4. Loop Filter Output

The loop filter has transfer function $F(s)$. Hence,

$$V_d(s) = F(s)V_e(s)$$

$$V_d(s) = K_d F(s)[\theta_i(s) - \theta_o(s)]$$

Here, $V_d(s)$ is the control voltage applied to the VCO.

5. VCO Transfer Function

For a VCO, the output angular frequency is controlled by the input voltage.

$$\omega_o(t) = \omega_{FR} + K_v V_d(t)$$

where,

$$\omega_{FR} = \text{free running angular frequency of VCO}$$

$$K_v = \text{VCO gain in rad/sec/volt}$$

Considering deviation from the free running frequency,

$$\Delta\omega_o(t) = K_v V_d(t)$$

But angular frequency is the derivative of phase:

$$\Delta\omega_o(t) = \frac{d\theta_o(t)}{dt}$$

Taking Laplace transform,

$$s\theta_o(s) = K_v V_d(s)$$

Therefore,

$$\theta_o(s) = \frac{K_v}{s} V_d(s)$$

Thus, the VCO behaves as an integrator with gain K_v , and its transfer function is

$$\boxed{\frac{\theta_o(s)}{V_d(s)} = \frac{K_v}{s}}$$

6. Closed Loop Transfer Function

From the PLL block diagram,

$$\theta_o(s) = \frac{K_v}{s} V_d(s)$$

Substituting $V_d(s) = K_d F(s)[\theta_i(s) - \theta_o(s)]$,

$$\theta_o(s) = \frac{K_v}{s} K_d F(s)[\theta_i(s) - \theta_o(s)]$$

Let,

$$K = K_d K_v$$

Then,

$$\theta_o(s) = \frac{KF(s)}{s} [\theta_i(s) - \theta_o(s)]$$

Multiplying by s ,

$$s\theta_o(s) = KF(s)\theta_i(s) - KF(s)\theta_o(s)$$

$$s\theta_o(s) + KF(s)\theta_o(s) = KF(s)\theta_i(s)$$

$$\theta_o(s)[s + KF(s)] = KF(s)\theta_i(s)$$

Therefore, the closed loop transfer function is

$$H(s) = \frac{\theta_o(s)}{\theta_i(s)} = \frac{KF(s)}{s + KF(s)}$$

where,

$$K = K_d K_v$$

7. Error Transfer Function

Since,

$$\phi(s) = \theta_i(s) - \theta_o(s)$$

$$\frac{\phi(s)}{\theta_i(s)} = 1 - \frac{\theta_o(s)}{\theta_i(s)}$$

$$\frac{\phi(s)}{\theta_i(s)} = 1 - \frac{KF(s)}{s + KF(s)}$$

$$\frac{\phi(s)}{\theta_i(s)} = \frac{s}{s + KF(s)}$$

Final Answer

Thus, the linear model of PLL gives the closed loop transfer function

$$\frac{\theta_o(s)}{\theta_i(s)} = \frac{KF(s)}{s + KF(s)}$$

and the phase error transfer function

$$\frac{\phi(s)}{\theta_i(s)} = \frac{s}{s + KF(s)}$$

where $K = K_d K_v$.

Solution to Question 2. (OR) (b)

Problem Statement: How can a Phase Locked Loop (PLL) be used as frequency demodulator?

Answer

A PLL can be used as a **frequency demodulator** because, in the locked condition, the VCO frequency follows the instantaneous frequency of the incoming FM signal. The control voltage of the VCO is therefore proportional to the instantaneous frequency deviation of the FM signal. Since frequency deviation is proportional to the original modulating signal, the loop filter output gives the demodulated signal.

1. Block Diagram of PLL as Frequency Demodulator

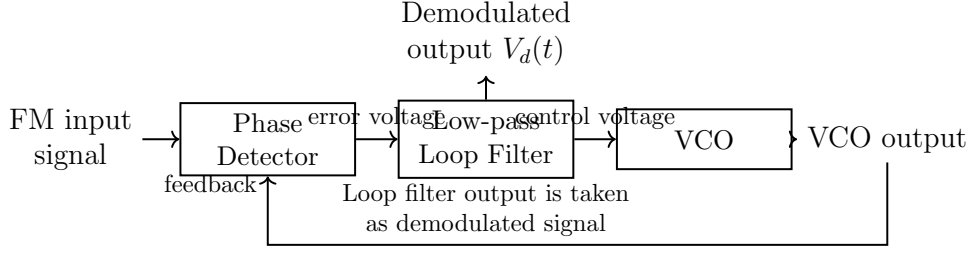


Figure 5: PLL used as frequency demodulator.

2. Frequency of FM Input Signal

Let the instantaneous angular frequency of the FM input signal be

$$\omega_i(t) = \omega_c + k_f m(t)$$

where,

$$\omega_c = \text{carrier angular frequency}$$

$$m(t) = \text{modulating signal}$$

$$k_f = \text{frequency sensitivity of FM modulator}$$

3. VCO Frequency

For the VCO,

$$\omega_o(t) = \omega_{FR} + K_v V_d(t)$$

where,

$$\omega_{FR} = \text{free running angular frequency of VCO}$$

$$K_v = \text{VCO sensitivity}$$

$$V_d(t) = \text{loop filter output or VCO control voltage}$$

For proper demodulation, the free running frequency of the VCO is adjusted equal to the carrier frequency:

$$\omega_{FR} = \omega_c$$

Therefore,

$$\omega_o(t) = \omega_c + K_v V_d(t)$$

4. Locked Condition

In the locked condition, the VCO frequency follows the input FM signal frequency.

$$\omega_o(t) = \omega_i(t)$$

So,

$$\omega_c + K_v V_d(t) = \omega_c + k_f m(t)$$

Cancelling ω_c from both sides,

$$K_v V_d(t) = k_f m(t)$$

Therefore,

$$V_d(t) = \frac{k_f}{K_v} m(t)$$

5. Explanation

The above equation shows that the loop filter output voltage $V_d(t)$ is directly proportional to the original modulating signal $m(t)$. Hence, the low-pass loop filter output of the PLL is the demodulated output.

$$V_{\text{demod}}(t) = V_d(t)$$

$$V_{\text{demod}}(t) \propto m(t)$$

6. Final Answer

A PLL acts as a frequency demodulator because its VCO tracks the instantaneous frequency of the incoming FM signal. The control voltage required to make the VCO frequency equal to the FM input frequency is proportional to the frequency deviation, and hence proportional to the original modulating signal.

Therefore,

$$V_{\text{demod}}(t) = V_d(t) = \frac{k_f}{K_v} m(t)$$

Thus, the loop filter output of the PLL gives the frequency demodulated signal.

Solution to Question 3

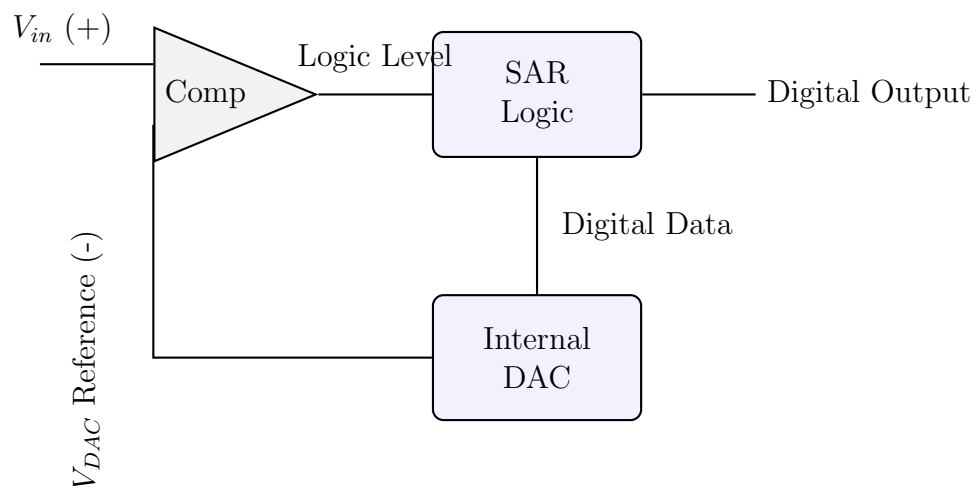
Solution to Question 3

3(a) Operation of Unipolar Successive Approximation ADC

The Successive Approximation Register (SAR) ADC converts an analog signal to digital using a binary search algorithm. It compares the input voltage (V_{in}) against the output of an internal DAC, testing each bit from the Most Significant Bit (MSB) to the Least Significant Bit (LSB).

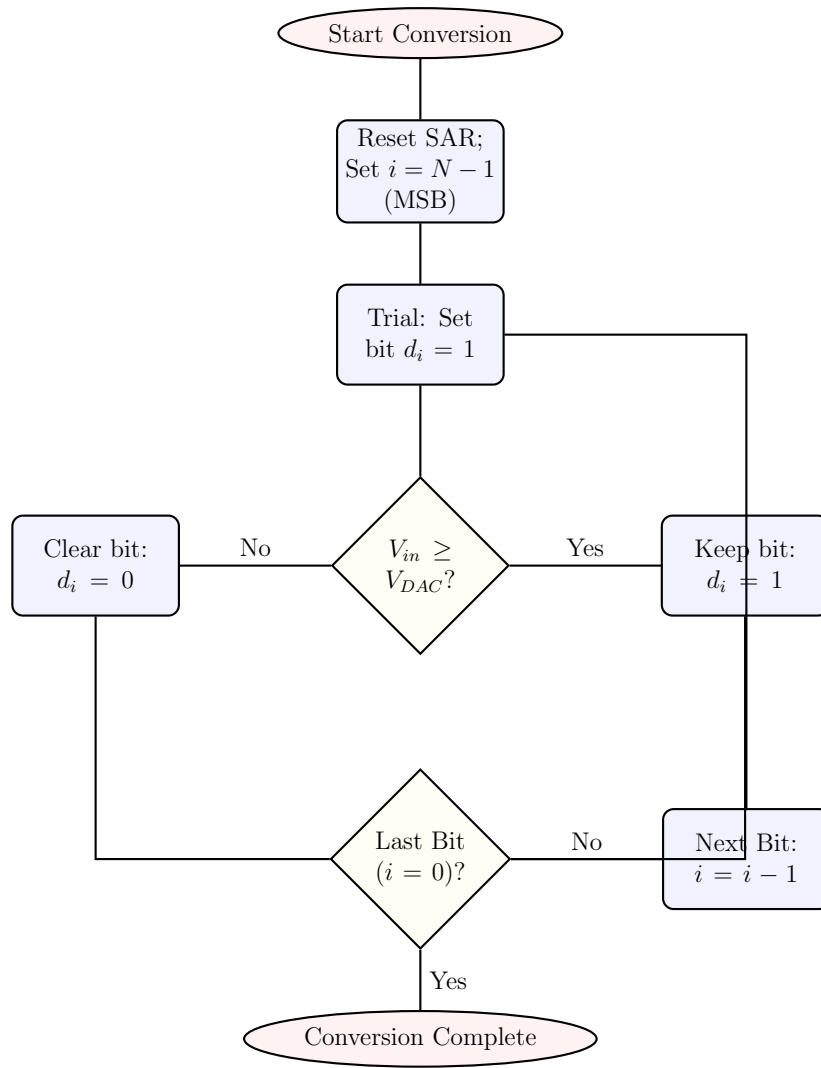
Block Diagram

In the diagram below, labels are placed with perpendicular offsets to ensure they do not overlap with the signal lines.



Flow Chart of the SAR Logic

The flow chart illustrates the "Set-and-Test" process used for each bit.



3(b) Calculation of 5-bit Binary Representation

Given Parameters:

- Input Voltage (V_{in}) = 10.317 V
- Reference Voltage (V_{ref}) = 16 V
- Resolution (N) = 5 bits

The value of the LSB is: $\Delta V = \frac{V_{ref}}{2^N} = \frac{16}{32} = 0.5$ V.

Successive Approximation Steps:

Step	Bit Weight	DAC Trial Voltage (V_{DAC})	Condition ($V_{in} \geq V_{DAC}$)	Result
1 (d_4)	8.0 V	8.0 V	$10.317 \geq 8.0$ (True)	1
2 (d_3)	4.0 V	$8.0 + 4.0 = 12.0$ V	$10.317 \geq 12.0$ (False)	0
3 (d_2)	2.0 V	$8.0 + 2.0 = 10.0$ V	$10.317 \geq 10.0$ (True)	1
4 (d_1)	1.0 V	$10.0 + 1.0 = 11.0$ V	$10.317 \geq 11.0$ (False)	0
5 (d_0)	0.5 V	$10.0 + 0.5 = 10.5$ V	$10.317 \geq 10.5$ (False)	0

Final Result: The 5-bit binary representation is **10100**.

3(c) Quantization Errors: Truncation vs. Rounding-off

Quantization error is the inherent uncertainty introduced when an analog signal is mapped to a finite digital level.

i. **Truncation Type ADC:** In truncation, the converter simply drops the fractional part of the signal beyond the LSB.

- **Error Range:** The error e_q is always between 0 and +1 LSB.
- **Impact:** This introduces a systematic ****DC bias**** or offset error of +1/2 LSB on average, which can degrade the accuracy of DC measurements.

ii. **Rounding-off Type ADC:** Rounding maps the analog value to the *nearest* digital level.

- **Error Range:** The error e_q is centered around zero, spanning $[-1/2 \text{ LSB}, +1/2 \text{ LSB}]$.
- **Impact:** This is generally preferred in signal processing because the ****average error is zero****. It behaves like white noise with a mean of zero, which can often be filtered out in AC applications.

Solution to Question 3. (OR) (a)

Problem Statement: Explain with circuit diagram the operation of an R-2R ladder Digital to Analog Converter (DAC) considering 3 bits.

Answer

An R-2R ladder DAC is a digital to analog converter which converts a digital binary input into a proportional analog output voltage. It uses only two values of resistors, namely R and $2R$. This is the main advantage of the R-2R ladder DAC over the binary-weighted resistor DAC, because in a binary-weighted DAC a large range of accurate resistor values is required.

For a 3-bit R-2R ladder DAC, let the digital input be

$$b_2b_1b_0$$

where,

$$b_2 = \text{MSB}, \quad b_1 = \text{middle bit}, \quad b_0 = \text{LSB}$$

Each bit controls one switch. If a bit is logic 1, the corresponding switch is connected to V_{ref} . If a bit is logic 0, the corresponding switch is connected to ground.

1. Circuit Diagram of 3-bit R-2R Ladder DAC

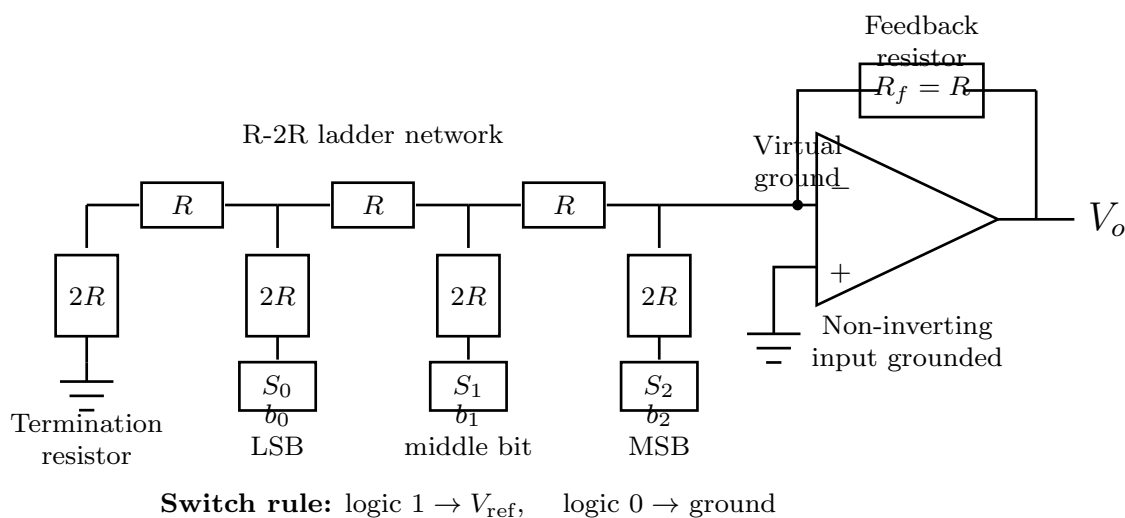


Figure 6: 3-bit R-2R ladder DAC using an inverting operational amplifier.

2. Operation of the Circuit

The R-2R ladder network divides the reference voltage in binary-weighted form. The important property of the ladder is that the resistance seen from every branch is $2R$.

Therefore, the current contribution of each successive bit becomes half of the previous bit.

For a 3-bit DAC:

$b_2 = \text{MSB}$ contributes $\frac{1}{2}$ of full-scale weight

$b_1 = \text{middle bit}$ contributes $\frac{1}{4}$ of full-scale weight

$b_0 = \text{LSB}$ contributes $\frac{1}{8}$ of full-scale weight

The op-amp is used in the inverting mode. Hence, the inverting terminal is at virtual ground. The binary-weighted currents from the ladder are summed at the inverting terminal of the op-amp and converted into an output voltage.

3. Output Voltage Equation

Let the reference voltage be V_{ref} . The bit values b_2 , b_1 , and b_0 can be either 0 or 1.

The total weighted contribution is

$$\frac{b_2}{2} + \frac{b_1}{4} + \frac{b_0}{8}$$

Therefore, for an inverting R-2R ladder DAC,

$$V_o = -\frac{R_f}{R} V_{\text{ref}} \left(\frac{b_2}{2} + \frac{b_1}{4} + \frac{b_0}{8} \right)$$

If $R_f = R$, then

$$\boxed{V_o = -V_{\text{ref}} \left(\frac{b_2}{2} + \frac{b_1}{4} + \frac{b_0}{8} \right)}$$

The negative sign appears because the op-amp is used in inverting configuration. If only magnitude is considered, then

$$\boxed{|V_o| = V_{\text{ref}} \left(\frac{b_2}{2} + \frac{b_1}{4} + \frac{b_0}{8} \right)}$$

4. Output Table for 3-bit R-2R DAC

Digital Input	Decimal Equivalent	$ V_o $
000	0	0
001	1	$\frac{V_{\text{ref}}}{8}$
010	2	$\frac{2V_{\text{ref}}}{8}$
011	3	$\frac{3V_{\text{ref}}}{8}$
100	4	$\frac{4V_{\text{ref}}}{8}$
101	5	$\frac{5V_{\text{ref}}}{8}$
110	6	$\frac{6V_{\text{ref}}}{8}$
111	7	$\frac{7V_{\text{ref}}}{8}$

5. Advantages of R-2R Ladder DAC

1. It uses only two resistor values, R and $2R$.
2. It is easier to fabricate in IC form.
3. It gives good accuracy compared with the binary-weighted resistor DAC.
4. It is suitable for higher number of bits.
5. It gives binary-weighted output voltage corresponding to the digital input.

Final Answer

Thus, a 3-bit R-2R ladder DAC converts the binary input $b_2b_1b_0$ into an analog output voltage given by

$$V_o = -V_{\text{ref}} \left(\frac{b_2}{2} + \frac{b_1}{4} + \frac{b_0}{8} \right)$$

for $R_f = R$.

Solution to Question 3. (OR) (b)

Problem Statement: Explain linearity error of a DAC.

Answer

In an ideal DAC, equal changes in the digital input code should produce equal changes in the analog output voltage. Therefore, the input-output characteristic of an ideal DAC should lie on a straight line.

But in a practical DAC, due to resistor tolerance, switch resistance, op-amp imperfections and reference voltage error, the actual output points may not lie exactly on the ideal straight line. This deviation is called **linearity error**.

1. Definition

Linearity error of a DAC is defined as the maximum deviation of the actual DAC output from the ideal straight-line transfer characteristic, after offset error and gain error have been adjusted to zero.

It is generally expressed in terms of LSB.

$$\text{Linearity Error} = \text{Actual output} - \text{Ideal straight-line output}$$

2. Diagram Showing Linearity Error

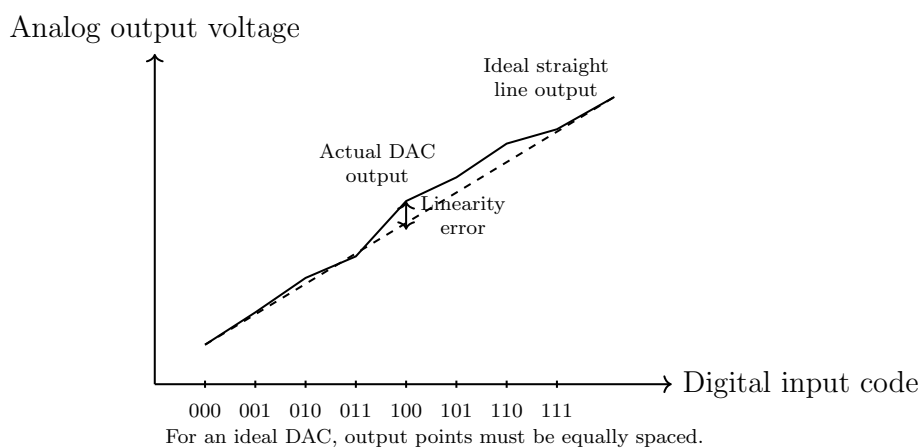


Figure 7: Linearity error in the transfer characteristic of a DAC.

3. Explanation

For an ideal DAC, the step size between two successive digital input codes is equal to one LSB.

$$1 \text{ LSB} = \frac{V_{\text{ref}}}{2^n}$$

where n is the number of bits.

If all output points are equally spaced, then the DAC is linear. But if the output voltage steps are not equal, the actual transfer characteristic bends away from the ideal straight line. This produces linearity error.

4. Important Points

1. Linearity error is measured after removing offset error and gain error.
2. It is the deviation of actual output from the ideal straight line.
3. It is usually expressed in LSB.
4. Large linearity error causes inaccurate analog output.
5. In severe cases, it may also lead to non-monotonic DAC behaviour.

Final Answer

Thus, DAC linearity error represents the deviation of the actual analog output from the ideal straight-line output characteristic. It is an important accuracy specification of a DAC and is generally expressed in LSB.

Solution to Question 3. (OR) (c)

Problem Statement: For a 4-bit DAC, the analog values corresponding to digital inputs 0000 and 0001 are 0.0125 V and 0.0625 V respectively. Find the output for an input 1110.

Answer

For a linear DAC, the difference between the output voltage for two successive digital codes is equal to one LSB step size.

Given,

$$V_{0000} = 0.0125 \text{ V}$$

$$V_{0001} = 0.0625 \text{ V}$$

1. Find One LSB Step Size

$$1 \text{ LSB} = V_{0001} - V_{0000}$$

$$1 \text{ LSB} = 0.0625 - 0.0125$$

$$\boxed{1 \text{ LSB} = 0.05 \text{ V}}$$

2. Convert Input 1110 to Decimal

The given input is

$$1110_2$$

$$1110_2 = 1 \times 2^3 + 1 \times 2^2 + 1 \times 2^1 + 0 \times 2^0$$

$$1110_2 = 8 + 4 + 2 + 0$$

$$\boxed{1110_2 = 14_{10}}$$

3. Calculate Output Voltage

Since the output for code 0000 is not zero, it acts as the offset output.

Therefore,

$$V_{\text{out}}(D) = V_{0000} + D(1 \text{ LSB})$$

For $D = 14$,

$$V_{\text{out}}(1110) = 0.0125 + 14(0.05)$$

$$V_{\text{out}}(1110) = 0.0125 + 0.7000$$

$$\boxed{V_{\text{out}}(1110) = 0.7125 \text{ V}}$$

Final Answer

Therefore, the output voltage for the digital input 1110 is

$$\boxed{0.7125 \text{ V}}$$

Solution to Question 4. (a)

Problem Statement: Show that the poles of Butterworth filter are located on an s -plane unit circle whereas Chebyshev poles are on an s -plane ellipse.

Answer

1. Butterworth Pole Location

For an n^{th} order normalized Butterworth low-pass filter, the magnitude squared function is

$$|H(j\omega)|^2 = \frac{1}{1 + \omega^{2n}}$$

For pole calculation, replace $j\omega$ by s . The poles of $H(s)H(-s)$ are obtained from

$$1 + (-1)^n s^{2n} = 0$$

This gives the $2n$ poles symmetrically placed on a circle. The stable Butterworth filter takes only the poles lying in the left half of the s -plane.

For the stable poles,

$$s_k = -\sin \theta_k + j \cos \theta_k$$

where,

$$\theta_k = \frac{(2k-1)\pi}{2n}, \quad k = 1, 2, 3, \dots, n$$

Now, let

$$s_k = \sigma_k + j\omega_k$$

So,

$$\sigma_k = -\sin \theta_k$$

and

$$\omega_k = \cos \theta_k$$

Therefore,

$$|s_k|^2 = \sigma_k^2 + \omega_k^2$$

$$|s_k|^2 = (-\sin \theta_k)^2 + (\cos \theta_k)^2$$

$$|s_k|^2 = \sin^2 \theta_k + \cos^2 \theta_k$$

$$|s_k|^2 = 1$$

Hence,

$$\boxed{|s_k| = 1}$$

Therefore, all normalized Butterworth poles lie on a circle of radius unity in the s -plane. The stable poles are selected from the left half of this unit circle.

2. Chebyshev Pole Location

For a Chebyshev filter, the magnitude squared function is

$$|H(j\omega)|^2 = \frac{1}{1 + \epsilon^2 T_n^2(\omega)}$$

where,

$$\epsilon = \text{ripple factor}$$

and

$$T_n(\omega) = \text{Chebyshev polynomial of order } n$$

The stable Chebyshev poles are given by

$$s_k = -\sinh \alpha \sin \theta_k + j \cosh \alpha \cos \theta_k$$

where,

$$\theta_k = \frac{(2k-1)\pi}{2n}$$

and

$$\alpha = \frac{1}{n} \sinh^{-1} \left(\frac{1}{\epsilon} \right)$$

Let

$$s_k = \sigma_k + j\omega_k$$

So,

$$\sigma_k = -\sinh \alpha \sin \theta_k$$

and

$$\omega_k = \cosh \alpha \cos \theta_k$$

Now,

$$\frac{\sigma_k}{\sinh \alpha} = -\sin \theta_k$$

and

$$\frac{\omega_k}{\cosh \alpha} = \cos \theta_k$$

Squaring and adding,

$$\left(\frac{\sigma_k}{\sinh \alpha}\right)^2 + \left(\frac{\omega_k}{\cosh \alpha}\right)^2 = \sin^2 \theta_k + \cos^2 \theta_k$$

$$\left(\frac{\sigma_k}{\sinh \alpha}\right)^2 + \left(\frac{\omega_k}{\cosh \alpha}\right)^2 = 1$$

Hence,

$$\boxed{\left(\frac{\sigma}{\sinh \alpha}\right)^2 + \left(\frac{\omega}{\cosh \alpha}\right)^2 = 1}$$

This is the equation of an ellipse. Therefore, Chebyshev poles lie on an ellipse in the s -plane. The stable Chebyshev poles are selected from the left half of this ellipse.

3. Pole Location Diagram

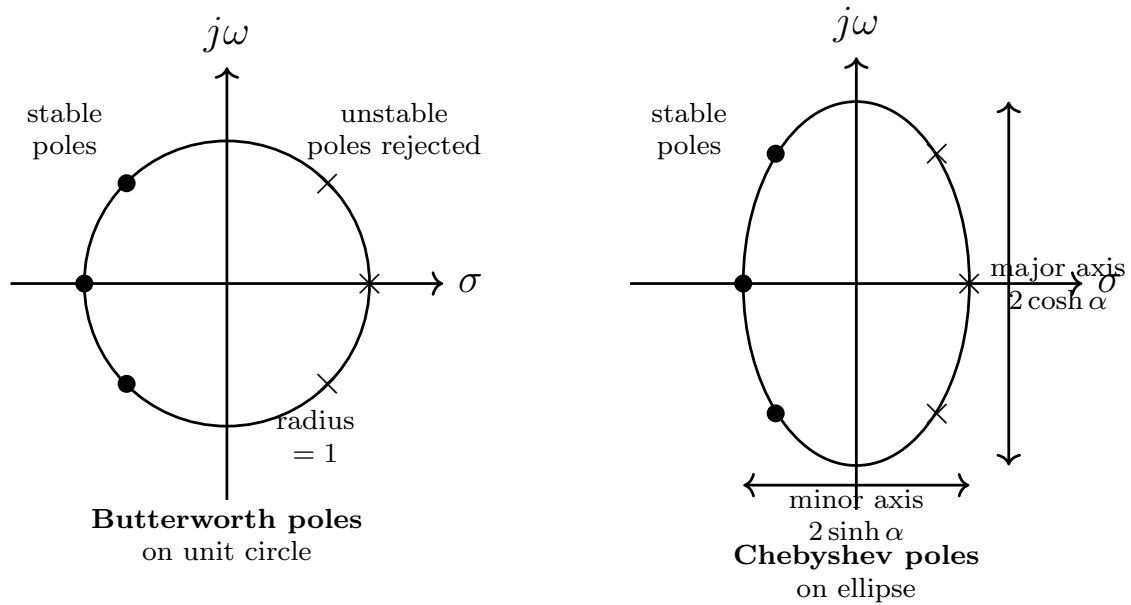


Figure 8: Butterworth poles lie on a unit circle, while Chebyshev poles lie on an ellipse.

Final Answer

Thus,

Butterworth poles lie on an s -plane unit circle

whereas

Chebyshev poles lie on an s -plane ellipse

Solution to Question 4. (b)

Problem Statement: Find the transfer function of a 2nd order band-pass filter circuit with center frequency $\omega_0 = 10 \text{ krad/sec}$, bandwidth $B = 2 \text{ krad/sec}$, and passband gain 10. Realize the band-pass filter circuit.

Answer

Given,

$$\omega_0 = 10 \text{ krad/sec} = 10 \times 10^3 \text{ rad/sec}$$

$$\boxed{\omega_0 = 10^4 \text{ rad/sec}}$$

and

$$B = 2 \text{ krad/sec} = 2 \times 10^3 \text{ rad/sec}$$

$$\boxed{B = 2000 \text{ rad/sec}}$$

Passband gain is

$$\boxed{A = 10}$$

1. Quality Factor

For a band-pass filter,

$$Q = \frac{\omega_0}{B}$$

Substituting values,

$$Q = \frac{10^4}{2 \times 10^3}$$

$$\boxed{Q = 5}$$

2. Standard Second Order Band-pass Transfer Function

The standard second order band-pass transfer function is

$$H_{BP}(s) = \frac{A B s}{s^2 + B s + \omega_0^2}$$

where,

A = passband gain

B = bandwidth

ω_0 = center frequency

Substituting,

$$A = 10, \quad B = 2000, \quad \omega_0 = 10^4$$

we get,

$$H_{BP}(s) = \frac{10(2000)s}{s^2 + 2000s + (10^4)^2}$$

$$H_{BP}(s) = \frac{20000s}{s^2 + 2000s + 10^8}$$

Hence,

$$H_{BP}(s) = \frac{20000s}{s^2 + 2000s + 10^8}$$

3. Verification of Passband Gain

At center frequency,

$$s = j\omega_0 = j10^4$$

Magnitude of band-pass transfer function at $\omega = \omega_0$ is

$$|H_{BP}(j\omega_0)| = \left| \frac{A B j\omega_0}{(j\omega_0)^2 + B j\omega_0 + \omega_0^2} \right|$$

Since,

$$(j\omega_0)^2 + \omega_0^2 = -\omega_0^2 + \omega_0^2 = 0$$

therefore,

$$|H_{BP}(j\omega_0)| = \left| \frac{A B j\omega_0}{B j\omega_0} \right|$$

$$|H_{BP}(j\omega_0)| = A = 10$$

Thus, the designed filter has passband gain 10.

4. State Variable Realisation of Band-pass Filter

A state variable realisation is convenient because it directly gives second order low-pass, high-pass and band-pass responses. For the state variable band-pass output,

$$H_{BP}(s) = \frac{-K_1 K_2 s}{s^2 + K_2 K_4 s + K_2 K_3}$$

For the required band-pass filter,

$$H_{BP}(s) = \frac{-20000s}{s^2 + 2000s + 10^8}$$

The negative sign only shows inversion due to the inverting integrator. The magnitude response is unchanged.

Comparing,

$$s^2 + K_2K_4s + K_2K_3 = s^2 + 2000s + 10^8$$

we get,

$$K_2K_3 = 10^8$$

and

$$K_2K_4 = 2000$$

Also,

$$K_1K_2 = 20000$$

Choose,

$$\boxed{K_2 = 10^4}$$

Then,

$$K_3 = \frac{10^8}{10^4} = 10^4$$

$$\boxed{K_3 = 10^4}$$

Also,

$$K_4 = \frac{2000}{10^4} = 0.2$$

$$\boxed{K_4 = 0.2}$$

and

$$K_1 = \frac{20000}{10^4} = 2$$

$$\boxed{K_1 = 2}$$

Therefore, one possible state variable realisation is

$$\boxed{K_1 = 2, \quad K_2 = 10^4, \quad K_3 = 10^4, \quad K_4 = 0.2}$$

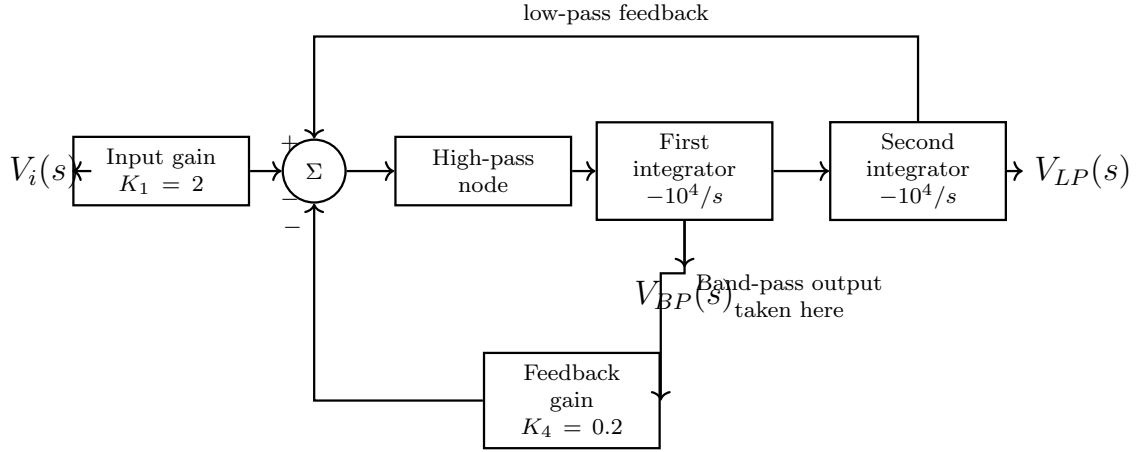


Figure 9: State variable realisation of the required second order band-pass filter.

5. Circuit Realisation Diagram

6. Practical Integrator Component Choice

For an op-amp integrator,

$$\frac{1}{RC} = 10^4$$

Choose,

$$C = 0.01 \mu\text{F} = 10^{-8} \text{ F}$$

Then,

$$R = \frac{1}{10^4 \times 10^{-8}}$$

$$R = 10^4 \Omega$$

$$\boxed{R = 10 \text{ k}\Omega}$$

So, each integrator may be realised using

$$\boxed{R = 10 \text{ k}\Omega, \quad C = 0.01 \mu\text{F}}$$

Final Answer

The required second order band-pass transfer function is

$$\boxed{H_{BP}(s) = \frac{20000s}{s^2 + 2000s + 10^8}}$$

and it has

$$\boxed{\omega_0 = 10^4 \text{ rad/sec}, \quad B = 2000 \text{ rad/sec}, \quad Q = 5}$$

Solution to Question 4. (c)

Problem Statement: Also realize the band-stop filter circuit from this band-pass filter having the same center frequency and passband gain.

Answer

A band-stop filter can be obtained from a band-pass filter by subtracting the band-pass output from the direct input path. Since the required passband gain is 10, the direct path gain is also taken as 10.

The band-pass transfer function obtained in Question 4(b) is

$$H_{BP}(s) = \frac{20000s}{s^2 + 2000s + 10^8}$$

The required band-stop transfer function with the same center frequency and passband gain is

$$H_{BS}(s) = 10 - H_{BP}(s)$$

Therefore,

$$H_{BS}(s) = 10 - \frac{20000s}{s^2 + 2000s + 10^8}$$

Taking LCM,

$$H_{BS}(s) = \frac{10(s^2 + 2000s + 10^8) - 20000s}{s^2 + 2000s + 10^8}$$

$$H_{BS}(s) = \frac{10s^2 + 20000s + 10^9 - 20000s}{s^2 + 2000s + 10^8}$$

$$H_{BS}(s) = \frac{10s^2 + 10^9}{s^2 + 2000s + 10^8}$$

$$H_{BS}(s) = \frac{10(s^2 + 10^8)}{s^2 + 2000s + 10^8}$$

1. Verification at Center Frequency

At the center frequency,

$$s = j\omega_0 = j10^4$$

The numerator becomes

$$10 [(j10^4)^2 + 10^8]$$

$$= 10[-10^8 + 10^8]$$

$$= 0$$

Therefore,

$$\boxed{H_{BS}(j\omega_0) = 0}$$

So the filter rejects the frequency $\omega_0 = 10^4 \text{ rad/sec}$.

2. Passband Gain Verification

At very low frequency,

$$s = 0$$

$$H_{BS}(0) = \frac{10(0 + 10^8)}{0 + 0 + 10^8}$$

$$\boxed{H_{BS}(0) = 10}$$

At very high frequency,

$$s \rightarrow \infty$$

$$H_{BS}(s) \approx \frac{10s^2}{s^2}$$

$$\boxed{H_{BS}(\infty) = 10}$$

Thus, the band-stop filter has passband gain 10 and the same center frequency as the band-pass filter.

3. Band-stop Realisation from Band-pass Filter

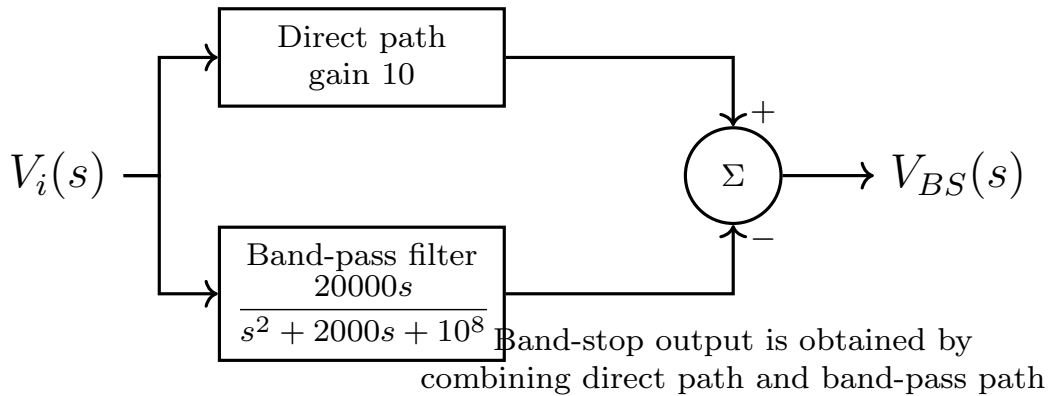


Figure 10: Band-stop filter obtained from the designed band-pass filter.

Final Answer

The required band-stop transfer function is

$$H_{BS}(s) = \frac{10(s^2 + 10^8)}{s^2 + 2000s + 10^8}$$

It has the same center frequency,

$$\omega_0 = 10^4 \text{ rad/sec}$$

and the same passband gain,

$$A = 10$$

Solution to Question 4. (OR) (a)

Problem Statement: Explain why the -3 dB cut-off frequency of a normalized Chebyshev filter is greater than 1 rad/sec.

Answer

For a normalized Chebyshev low-pass filter, the magnitude squared response is

$$|H(j\omega)|^2 = \frac{1}{1 + \epsilon^2 T_n^2(\omega)}$$

where,

ϵ = ripple factor

and

$T_n(\omega)$ = Chebyshev polynomial of order n

For a normalized Chebyshev filter, the frequency

$$\omega = 1 \text{ rad/sec}$$

does not represent the -3 dB cut-off frequency. Instead, it represents the **end of the pass-band ripple region**.

1. Magnitude at $\omega = 1$

For Chebyshev polynomial,

$$T_n(1) = 1$$

Therefore,

$$|H(j1)|^2 = \frac{1}{1 + \epsilon^2}$$

Now, -3 dB cut-off means half-power frequency. Hence,

$$|H(j\omega_{3dB})|^2 = \frac{1}{2}$$

At $\omega = 1$,

$$|H(j1)|^2 = \frac{1}{1 + \epsilon^2}$$

For usual Chebyshev filters, the pass-band ripple is less than 3 dB. Therefore,

$$\epsilon < 1$$

So,

$$1 + \epsilon^2 < 2$$

Thus,

$$\frac{1}{1 + \epsilon^2} > \frac{1}{2}$$

Hence, at $\omega = 1$, the magnitude has not yet fallen to the half-power value. Therefore,

$$\boxed{\omega_{3dB} > 1 \text{ rad/sec}}$$

2. Mathematical Derivation of -3 dB Frequency

At -3 dB frequency,

$$|H(j\omega_{3dB})|^2 = \frac{1}{2}$$

So,

$$\frac{1}{1 + \epsilon^2 T_n^2(\omega_{3dB})} = \frac{1}{2}$$

Cross multiplying,

$$1 + \epsilon^2 T_n^2(\omega_{3dB}) = 2$$

$$\epsilon^2 T_n^2(\omega_{3dB}) = 1$$

$$T_n(\omega_{3dB}) = \frac{1}{\epsilon}$$

For $\omega > 1$,

$$T_n(\omega) = \cosh [n \cosh^{-1}(\omega)]$$

Therefore,

$$\cosh [n \cosh^{-1}(\omega_{3dB})] = \frac{1}{\epsilon}$$

Taking inverse hyperbolic cosine,

$$n \cosh^{-1}(\omega_{3dB}) = \cosh^{-1} \left(\frac{1}{\epsilon} \right)$$

$$\cosh^{-1}(\omega_{3dB}) = \frac{1}{n} \cosh^{-1} \left(\frac{1}{\epsilon} \right)$$

Hence,

$$\boxed{\omega_{3dB} = \cosh \left[\frac{1}{n} \cosh^{-1} \left(\frac{1}{\epsilon} \right) \right]}$$

Since $\epsilon < 1$,

$$\frac{1}{\epsilon} > 1$$

Therefore,

$$\omega_{3dB} > 1$$

3. Diagram

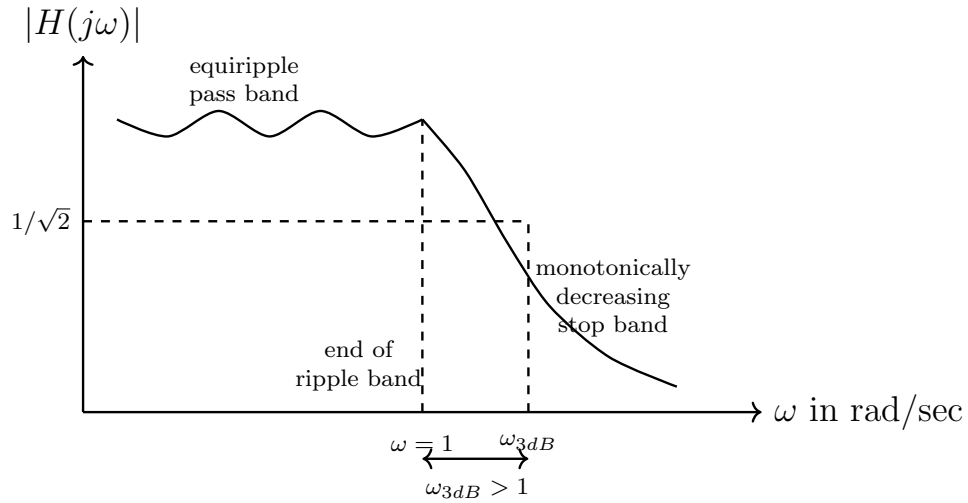


Figure 11: In a normalized Chebyshev filter, $\omega = 1$ is the end of ripple band, while ω_{3dB} occurs later.

Final Answer

Thus, in a normalized Chebyshev filter, $\omega = 1$ rad/sec represents the end of the ripple band. The -3 dB frequency occurs when the magnitude becomes $1/\sqrt{2}$, and for ordinary ripple values $\epsilon < 1$, this happens at a frequency greater than 1 rad/sec.

$\omega_{3dB} > 1 \text{ rad/sec}$

Solution to Question 4. (OR) (b)

Problem Statement: Find out the minimum order and ripple factor of a normalized Chebyshev low-pass filter with maximum passband ripple attenuation 0.3 dB and stop-band attenuation minimum 40 dB for $\omega \geq 10$ rad/sec.

Answer

Given,

$$A_{\max} = 0.3 \text{ dB}$$

$$A_{\min} = 40 \text{ dB}$$

$$\omega_p = 1 \text{ rad/sec}$$

because the filter is normalized, and

$$\omega_s = 10 \text{ rad/sec}$$

1. Ripple Factor

For Chebyshev filter,

$$A_{\max} = 10 \log_{10}(1 + \epsilon^2)$$

Therefore,

$$1 + \epsilon^2 = 10^{A_{\max}/10}$$

$$\epsilon^2 = 10^{A_{\max}/10} - 1$$

$$\epsilon = \sqrt{10^{A_{\max}/10} - 1}$$

Substituting,

$$\epsilon = \sqrt{10^{0.3/10} - 1}$$

$$\epsilon = \sqrt{10^{0.03} - 1}$$

$$\epsilon = \sqrt{1.07152 - 1}$$

$$\epsilon = \sqrt{0.07152}$$

$$\boxed{\epsilon = 0.2674}$$

2. Minimum Order

For Chebyshev low-pass filter,

$$n \geq \frac{\cosh^{-1} \left[\sqrt{\frac{10^{A_{\min}/10}-1}{10^{A_{\max}/10}-1}} \right]}{\cosh^{-1} \left(\frac{\omega_s}{\omega_p} \right)}$$

Substituting,

$$n \geq \frac{\cosh^{-1} \left[\sqrt{\frac{10^{40/10}-1}{10^{0.3/10}-1}} \right]}{\cosh^{-1} \left(\frac{10}{1} \right)}$$

$$n \geq \frac{\cosh^{-1} \left[\sqrt{\frac{10^4-1}{0.07152}} \right]}{\cosh^{-1}(10)}$$

$$n \geq \frac{\cosh^{-1} \left[\sqrt{\frac{9999}{0.07152}} \right]}{\cosh^{-1}(10)}$$

$$n \geq \frac{\cosh^{-1}(373.91)}{\cosh^{-1}(10)}$$

Now,

$$\cosh^{-1}(373.91) = 6.617$$

and

$$\cosh^{-1}(10) = 2.993$$

Therefore,

$$n \geq \frac{6.617}{2.993}$$

$$n \geq 2.21$$

Since the order must be an integer,

$$\boxed{n = 3}$$

Final Answer

The ripple factor is

$$\boxed{\epsilon = 0.2674}$$

and the minimum order of the normalized Chebyshev low-pass filter is

$$\boxed{n = 3}$$

Solution to Question 4. (OR) (c)

Problem Statement: Design a low-pass filter with the following specifications using any known filter topology:

- 3 dB cut-off frequency = 2 kHz.
 - Pass-band attenuation is maximum 0.5 dB for $f \leq 1$ kHz.
 - Stop-band attenuation is minimum 10 dB for $f \geq 5$ kHz.
 - Pass-band gain is 2.5.
-

Answer

We shall design a **Butterworth low-pass filter**, because Butterworth response is maximally flat in the pass-band and the 3 dB cut-off frequency is directly specified.

1. Given Specifications

$$f_c = 2 \text{ kHz}$$

$$f_p = 1 \text{ kHz}$$

$$f_s = 5 \text{ kHz}$$

$$A_p = 0.5 \text{ dB}$$

$$A_s = 10 \text{ dB}$$

$$K = 2.5$$

The angular cut-off frequency is

$$\omega_c = 2\pi f_c$$

$$\omega_c = 2\pi(2000)$$

$$\boxed{\omega_c = 4000\pi = 12566 \text{ rad/sec}}$$

2. Find the Minimum Order

For a Butterworth low-pass filter,

$$A(\omega) = 10 \log_{10} \left[1 + \left(\frac{\omega}{\omega_c} \right)^{2n} \right]$$

At pass-band edge,

$$\frac{f_p}{f_c} = \frac{1}{2} = 0.5$$

At stop-band edge,

$$\frac{f_s}{f_c} = \frac{5}{2} = 2.5$$

Check for $n = 1$

At $f_p = 1$ kHz,

$$A_p = 10 \log_{10} [1 + (0.5)^2]$$

$$A_p = 10 \log_{10}(1.25)$$

$$A_p = 0.969 \text{ dB}$$

This is greater than 0.5 dB, so $n = 1$ is not acceptable.

At $f_s = 5$ kHz,

$$A_s = 10 \log_{10} [1 + (2.5)^2]$$

$$A_s = 10 \log_{10}(7.25)$$

$$A_s = 8.60 \text{ dB}$$

This is less than 10 dB, so $n = 1$ is not acceptable.

Check for $n = 2$

At $f_p = 1$ kHz,

$$A_p = 10 \log_{10} [1 + (0.5)^4]$$

$$A_p = 10 \log_{10}(1.0625)$$

$$A_p = 0.263 \text{ dB}$$

This is less than 0.5 dB, so the pass-band condition is satisfied.

At $f_s = 5$ kHz,

$$A_s = 10 \log_{10} [1 + (2.5)^4]$$

$$A_s = 10 \log_{10}(40.0625)$$

$$A_s = 16.03 \text{ dB}$$

This is greater than 10 dB, so the stop-band condition is also satisfied.
Therefore,

$$\boxed{n = 2}$$

3. Transfer Function

The normalized second order Butterworth low-pass transfer function is

$$H_n(s) = \frac{1}{s^2 + \sqrt{2}s + 1}$$

After frequency scaling,

$$s \rightarrow \frac{s}{\omega_c}$$

So,

$$H(s) = \frac{K\omega_c^2}{s^2 + \sqrt{2}\omega_c s + \omega_c^2}$$

Since $K = 2.5$,

$$H(s) = \frac{2.5\omega_c^2}{s^2 + \sqrt{2}\omega_c s + \omega_c^2}$$

Putting,

$$\omega_c = 12566 \text{ rad/sec}$$

we get,

$$\omega_c^2 = (12566)^2 = 1.579 \times 10^8$$

$$\sqrt{2}\omega_c = 1.414(12566) = 17772$$

and

$$2.5\omega_c^2 = 3.948 \times 10^8$$

Therefore,

$$\boxed{H(s) = \frac{3.948 \times 10^8}{s^2 + 1.7772 \times 10^4 s + 1.579 \times 10^8}}$$

4. Sallen-Key Realisation

We shall realise the second-order Butterworth low-pass section using a unity-gain Sallen-Key low-pass filter and then add a non-inverting amplifier of gain 2.5.

For normalized second order Butterworth response,

$$Q = \frac{1}{\sqrt{2}} = 0.707$$

One convenient normalized Sallen-Key choice is

$$R_1 = R_2 = 1 \Omega$$

$$C_3 = 2Q = \sqrt{2} \text{ F}$$

$$C_4 = \frac{1}{2Q} = \frac{1}{\sqrt{2}} \text{ F}$$

Thus,

$$C_3 = 1.414 \text{ F}$$

$$C_4 = 0.707 \text{ F}$$

5. Frequency and Impedance Scaling

The required cut-off frequency is

$$\omega_c = 12566 \text{ rad/sec}$$

After frequency scaling,

$$R'_1 = R'_2 = \frac{1}{\omega_c}$$

$$R'_1 = R'_2 = \frac{1}{12566}$$

$$R'_1 = R'_2 = 7.958 \times 10^{-5} \Omega$$

Now choose a practical capacitor value

$$C''_4 = 0.01 \mu\text{F}$$

The old normalized value was

$$C_4 = 0.707 \text{ F}$$

So the impedance scaling ratio is

$$m = \frac{C''_4}{C_4}$$

$$m = \frac{0.01 \times 10^{-6}}{0.707}$$

$$m = 1.414 \times 10^{-8}$$

Therefore,

$$R_1'' = R_2'' = \frac{R'}{m}$$

$$R_1'' = R_2'' = \frac{7.958 \times 10^{-5}}{1.414 \times 10^{-8}}$$

$$\boxed{R_1'' = R_2'' = 5.63 \text{ k}\Omega}$$

Also,

$$C_3'' = mC_3$$

$$C_3'' = (1.414 \times 10^{-8})(1.414)$$

$$\boxed{C_3'' = 0.02 \mu\text{F}}$$

and

$$\boxed{C_4'' = 0.01 \mu\text{F}}$$

6. Gain Stage

The Sallen-Key section is taken as unity gain. To obtain pass-band gain 2.5, a non-inverting amplifier is connected after the filter.

For non-inverting amplifier,

$$K = 1 + \frac{R_f}{R_g}$$

We need,

$$K = 2.5$$

So,

$$2.5 = 1 + \frac{R_f}{R_g}$$

$$\frac{R_f}{R_g} = 1.5$$

Choose,

$$R_g = 10 \text{ k}\Omega$$

$$R_f = 15 \text{ k}\Omega$$

Therefore,

$$K = 1 + \frac{15}{10} = 2.5$$

7. Complete Circuit Diagram

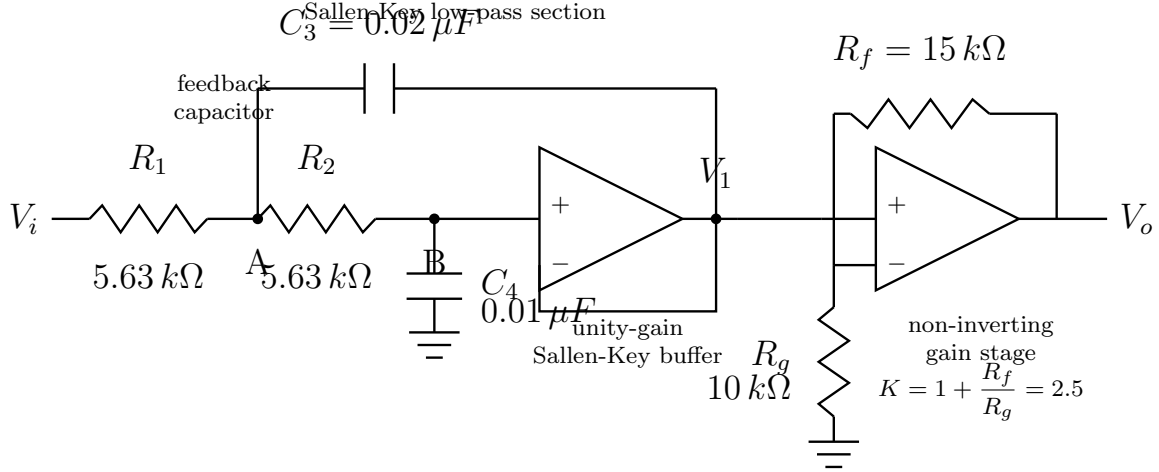


Figure 12: Corrected Sallen-Key low-pass circuit followed by a non-inverting gain stage of 2.5.

8. Final Design Values

The designed filter is a second order Butterworth low-pass filter with

$$n = 2$$

$$\omega_c = 12566 \text{ rad/sec}$$

$$H(s) = \frac{3.948 \times 10^8}{s^2 + 1.7772 \times 10^4 s + 1.579 \times 10^8}$$

One possible practical realization is:

$$R_1 = R_2 = 5.63 \text{ k}\Omega$$

$$C_3 = 0.02 \mu\text{F}$$

$$C_4 = 0.01 \mu\text{F}$$

and the pass-band gain stage uses

$$R_g = 10 \text{ k}\Omega, \quad R_f = 15 \text{ k}\Omega$$

So the total pass-band gain is

$$K = 2.5$$